Prop. If $p>0$, then

$$\lim_{n\to\infty} \frac{1}{n^p} = 0 \quad \text{and} \quad \lim_{n\to\infty} \sqrt[n]{p} = 1.$$

proof. Given $\varepsilon>0$, take $N \in \mathbb{N}$ s.t. $N > \frac{1}{\sqrt[p]{\varepsilon}}$. Then

$$n \geq N \Rightarrow \frac{1}{n^p} \leq \frac{1}{N^p} < \varepsilon$$

And, if $p=1$, directly $\sqrt[n]{p} = 1 \to 1$.

If $p>1$, let $x_n = \sqrt[n]{p} - 1$. Then, $x_n > 0$, and

$$p = (x_n+1)^n \geq n x_n \Rightarrow x_n \leq \frac{p}{n}, \text{ so } x_n \to 0 \text{ as } n\to\infty.$$

If $p<1$, then $\sqrt[n]{\frac{1}{p}} \to 1$, so $\sqrt[n]{p} \to 1$.

Prop. $\lim_{n\to\infty} \sqrt[n]{n} = 1$.

proof. Let $x_n = \sqrt[n]{n} - 1$.